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Quality of Life Considerations During Cancer Treatment in Invasive Ductal Carcinoma Patients: A Systemic Review

Abstract: *Breast cancer is the number two leading cause of death in women all over the world and is often associated with poor quality of life (QOL). The positive and negative QOL factors influence the overall health and well-being of those affected with invasive ductal carcinoma (IDC). This literature review was structured to identify and understand both the positive and negative QOL factors throughout breast cancer treatment, as well as post breast cancer treatment. Systemic searches were done of three databases to gather data in breast cancer treatment from 2010-2015. Results identified the positive and negative factors associated with the QOL in relation to breast cancer treatment. The most prevalent positive QOL factors included patient expectations, decreased side effects, and increased survival rate. The most prevalent negative QOL factors included treatment, specific side effects and decreased quality of life. This review may guide healthcare professionals in incorporating new practices and identifying the best regimen to improving QOL. The positive and negative QOL factors, in relation to treatment, are important because they help healthcare professionals understand how those factors impact the overall health and well-being of individuals with IDC.*

Key Words: *Invasive Ductal Carcinoma, Quality of Life, Treatment*

Introduction

Cancer is second only to heart disease as a leading cause of death among all ages of women and men (Siegel, Miller, & Jemal, 2015). Breast cancer itself is the number two cause of death among cancers in women (Siegel, et al., 2014). Breast cancer incidence is often associated with poor quality of life (QOL) for those who are suffering (Hershman, Kushi, Hillyer, Coromilis, Buono, Lamerato, Bovbjerd, Mandelblatt, Tsai, Jacobson, Wright, & Neugut, 2016; Howard-Anderson, Ganz, Bower, & Stanton, 2012; Jeste, Liu, Rissling, Trofimenko, and Natarajan, Parker, & Ancoli-Israel, 2013). The World Health Organization (WHO) defines quality of life as "an individual's perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards, and concerns" (WHO, 2016). QOL can compare between two levels: the objective level and the subjective level. Objectivity refers to the tangible components that make up the basic level of QOL, which includes needs such as air, food, and environment. Subjectivity refers to the intangible aspects of QOL, which includes the level of satisfaction with which individuals respond to the respective QOL factors (Bulley, Gaal, Coutts, Blyth, Jack, Chetty, Barber, & Tan, 2013).

Patients go through intense experiences that affect their daily living while being treated for breast cancer (Young, Weltzien, Kwan, Castillo, Caan, & Kroenke, 2014). The purpose of this systematic review was to identify and understand both the positive and negative QOL factors throughout breast cancer treatment as well as post breast cancer treatment. The information gathered from this study may be applied to medical practices in order to increase the QOL for metastatic invasive ductal carcinoma cancer (IDC) patients.

The National Cancer Institute (NCI) (2013) defines metastases as "the spread of cancer from one part of the body to another." The metastatic process includes the cancer cells breaking away from their original location, traveling through areas of the body such as blood and lymph, and forming new cancer cells (tumors) in different parts of the body (Geetha & Selvi, 2015). The newly formed tumors are known as metastatic tumors which contain similar cellular components found in the original cancer cells (Chou & Friedman, 2016). IDC is the most common type of invasive breast cancer (Carbognin, Sperduti, Brunelli, Marcolini, Nortilli, Pilotto, Zampiva, Merler, Filrio, Filippi, Manfrin, Pellini, Bonette, Pollini, Tortora, & Bria, 2016). IDC develops in the lining of the milk ducts and metastasizes to the surrounding tissue (Hinck & Nathke, 2014). Furthermore, IDC can metastasize through the blood and lymphatic systems to other parts of the body (NCI,2013).

The findings of this systematic literature review may also be useful for physicians (oncologists in particular), as they discuss the course of treatment with their IDC patients. Physicians will be able to identify the best practices to implement as is fitting for the patient's preferences. Other clinicians, such as nurses, may find this review assistive in the daily care of the patients in the hospital. In addition, healthcare professionals will be able to create and execute policies that focus towards IDC patients and the increase of their QOL. Data for this review were gathered using PubMed (queries

MEDLINE), Academic Search Complete, and Cumulative Index of Nursing and Allied Health Literature (CINAHL). The search criteria focused on QOL and treatment, with respect to metastasis of IDC. Articles based on positive and negative QOL factors, and the data were summarized.

SAMPLE

Research databases were queried from PubMed, Academic Search Complete, and CINAHL using search terms "("invasive ductal carcinoma" OR "ductal carcinoma") AND (treatment OR medication) AND quality of life AND metastasis." Several exclusion criteria were also specified: duplicates were excluded that were non-germane articles, and articles not available in English. Searches were limited to academic journals and articles in English language from 2010-2015 (n=9). (See Table 1).

During the five-year period chosen (2010-2015), 53 authors published 9 works that specifically studied, analyzed, and discussed IDC in relation to quality of life, treatment, approach to care, physical/mental health, and metastasis. Most of these works highlighted both positive and negative QOL factors, however some focused on positive QOL factors only. Articles originated from multiple countries, and a total of 45 positive QOL factors (26, or 58%) and negative QOL factors (19, or 42%) were observed.

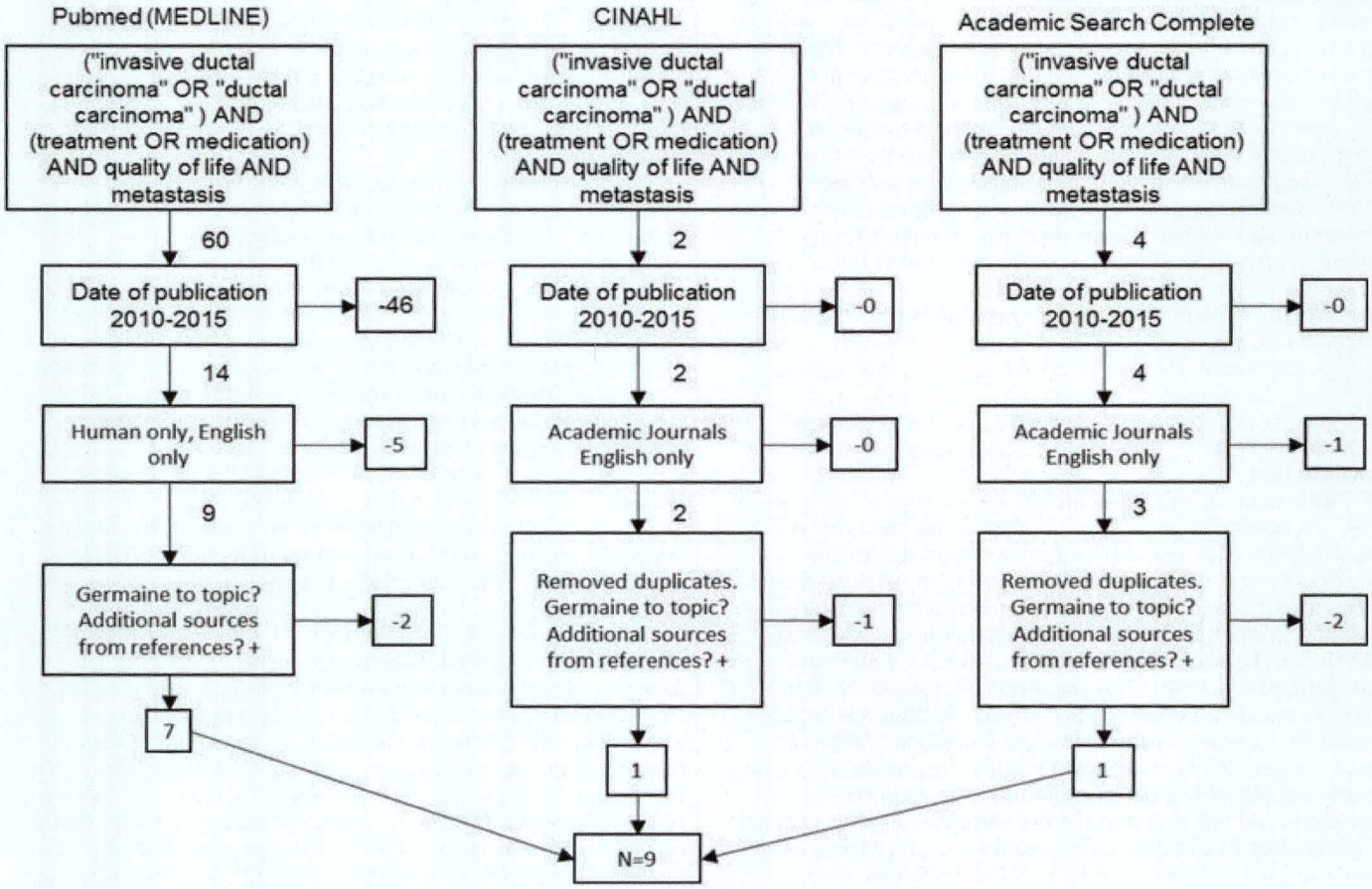
DISCUSSION

About 27% more positive QOL factors were noted than negative ones (26:19). The authors in this review compared and grouped the positive and negative QOL factors into common themes. A total of 18 themes were noted between the two categories. There were 12 positive QOL categories identified, and 6 negative QOL categories identified. Illustrated in rank, the themes from the positive QOL factors are based on frequency of their occurrence in the literature.

The theme surrounding positive QOL factors mentioned most often in the literature was tied between patient expectations (Fenicia, et al., 2014; Ghirarduzzi, Sivelli, Martella, Bella, De Simone, Arcuri, Zannoni, Del Rio, & Sianesi, 2010; Hanna, de Barros, de Andrade, Bevilacqua, Piato, Pelosi, Martella, da Silva, & Carvalho, 2014; Ju, et al., 2013; and Kechagias, Ntomouchtsis, Patrikidou, Christoforidou, Andreadis, & Vahtsevanos, 2012). Decreased side effects were noted by Dennis & Chow, 2011; Fenicia, Abdolrahimzadeh, Mannino, Verrilli, Balestrieri, & Recupero, 2014; Hanna, et al., 2014; Ju, et al., 2013; and Rancati, Soderini, Dorr, Gercovich, Tessari, & Gonzalez, 2013). These themes were identified 5 times each, out of 26 total occurrences (19% each). Regarding patient expectations, the QOL areas where there were various improvements in patients, allowed for higher patient expectations. These areas were surrounding cosmetic reconstruction outcomes, visual improvements in patients after metastasis, updated dosing schedules of chemotoxic drugs, and disease management being in the context of a treatment itself. Each of these areas were found to be ways that would allow for more positive QOLs for patients if they were addressed adequately, as many of them were not really a part of the treatment protocol for IDC up until they were individually studied.

Decreased side effects was the other most cited positive QOL factor. These articles focused on treatment factors which would decrease incidence of side effects and complications, overall. Changing treatment protocols, or utilizing adjunct

Table 1. The members of the research team reevaluated the articles and determined positive and negative factors affecting metastasis in IDC patients



therapies alongside traditional ones, allowed for decreased side effects in some patients. Understanding that many problems in treatment of IDC come from the side effects of the treatment itself, and working to mitigate these factors was the focus of the articles.

Tied for 12% each (3 articles) were the themes of increased survival rate (Bianchi, Duca, Sica, & Mariani, 2012; Ghirarduzzi, et al., 2010; and Ju, et al., 2013) and ease of treatment (Dennis & Chow, 2011; and Fenicia, et al., 2014). Increased survival rate speaks for itself in being a positive QOL factor. These articles were focused on utilization of specific protocols and interventions which resulted in a longer lifespan for IDC patients. Ease of treatment conversely was focused on finding alternative treatments which made the traditional ones much less of a burden on patients. Decreasing treatment time, treatment number, increasing non-invasive measures, and avoiding invasive procedures allowed for a much easier treatment plan for patients.

Two other factors also were tied in two other articles among the themes at 8% each. These themes were the decrease in metastasis (Hanna, et al., 2014; and Ju, et al., 2013) and emotional advantage (Asevedo, Brietzke, & Chaves, 2013; and Rancati, et al., 2013). Decreased metastasis themed articles observed that targeted therapies could indeed decrease the incidence of distant metastasis. Emotional advantages were noted as positive QOL factors. These were tools such as psychotherapy and cosmetic procedures that could be used to assist in psychological issues resulting from IDC treatment. Addressing these issues allowed for a more positive treatment plan for patients.

Lastly, six articles were tied in the last six themes, each of them being cited in the literature only one time (each accounting for 4%). The six themes were cost effective (Hanna, et al., 2014), decreased occurrence (Fenicia, Abdolrahimzadeh, Mannino, Verrilli, Balestrieri, & Recupero, 2014), decreased recurrence increased aesthetics multiple treatment success (Ju, et al., 2013), and patient satisfaction increased (Rancati, et al., 2013). Researchers found that alternative therapies were more cost effective, with increased results, over traditional beam therapies in radiation therapy. Ways to decrease occurrence of tumors surrounded methods which were being used to cause specific tumor regression. Decreased recurrence of tumors was shown to be significantly affected after tumor specific radiation therapies were applied to the area. Increased aesthetics allowed for not only positive QOL factors, but also increased positive mental health for IDC patients, as well. Multiple treatment successes were seen where patients began to utilize a mixture of different therapies after radiotherapies and chemotherapies (traditional therapies) failed. Patient satisfaction was found to be increased when breast reconstruction occurred, along with additional psychological benefits as well.

The negative QOL factors were not as numerous in the sample of 9 articles chosen for this review. A total of six themes were identified over 19 occurrences. Illustrating and ranking orders the themes from the negative QOL factors were based on frequency of their occurrence in the literature.

By a large margin, the theme treatment specific side effects were mentioned the most often with 9 occurrences out of a total of 19 (or 47%) (Asevedo, et al., 2013; Dennis & Chow, 2011; Ju, et al., 2013; Kechagias, et al., 2012; and Rancati, et al., 2013). This theme encompassed observations that certain treatments had very specific side effects associated with them that could cause suffering in patients. These side effects included treatment site fibrosis, headaches, neurological symptoms, seizures, strokes, cognitive dysfunction, negative aesthetics, bone pain, fatigue, anorexia, constipation, medication side effects, and other negative side effects from differing treatment modalities. Any or all of these side effects may lead to a negative QOL situation for patients suffering from IDC.

Decreased quality of life was the negative factor mentioned in 3 of 19 occurrences (18%) (Asevedo, et al., 2013; Dennis & Chow, 2011; and Kechagias, et al., 2012). Authors also noted problems with mood swings, sleep disturbances, hallucinations, memory loss, seizures, metastases, and other symptoms which were problematic for patients. Any one of these areas could quickly amount to a QOL change for patients. In many instances these issues are compounded with each other, or exhibit themselves in specific triads of symptoms due to treatments (such as swelling, teeth mobility, and trismus).

Two barriers were mentioned most often, which were ineligible for treatment (Hanna, et al., 2014; and Rancati, et al., 2013) and decreased life span (Dennis & Chow, 2011; and Ju, et al., 2013; Each of these were mentioned in 2 of 19 occurrences (11%). It was noted that numerous patients were ineligible for treatment due to not being able to take part in study groups pertaining to their specific type of cancer. This inability was found to be rooted in these patients not being suitable for certain study groups and therefore, not being able to benefit from them such as smokers were unsuitable for certain study groups. Decreased life span was noted in those who had metastasis. The metastasis of cancer to other areas of the body from the breast was often a life-limiting factor as it had an overall poor prognosis.

The last two barriers found were psychological side effects (Asevedo, et al., 2013) and no increase to survival rates (Bianchi, Duca, Sica, & Mariani, 2012). Each of these factors were found in only one of the articles, yielding a 5% occurrence for each. Psychological side effects were significant because as many as 50% of patients with cancer also had a concomitant mental disorder, as well. Also found was no increase to survival rates with the use of some types of therapy protocols.

This systematic review focused on several areas of QOL including treatment, approach to care, physical/mental health, and metastasis. The course of treatment and all its aspects such as the aggressiveness and the length of each treatment can affect QOL in which much of a patient's daily life can be spent undergoing treatment and physical pain. Additionally, the effects of the treatments can have effects on physical, emotional, and mental QOL. Patients may experience increased fatigue, weakness, depression, and anxiety. These issues may also be a factor as to a negative approach to care. Therefore, because each patient responds differently to treatment, it is important for physicians to thoroughly assess options individually for each patient before a treatment regimen has been chosen.

APPROACH TO CARE

The approach to care differs between patients according to the severity of their disease. Rancati, et al., 2013 suggested that psychological counseling and appropriate therapy induced side effects management should be highly considered for the care of metastatic IDC. In addition, a few authors (Dennis & Chow, 2011; and Kechagias, et al., 2012) have also suggested aggressive management for patients diagnosed with IDC. "Management in such cases should be in the context of the treatment of a metastatic disease that could prolong survival and improve quality of life" (Kechagias, et al., 2012). Many of the literary articles supported the success of breast cancer treatment. Depending on the stage of cancer, treatments to destroy or slow the growth of the tumor are administered, such as surgery, chemotherapy, and hormone and radiation therapy (Dennis & Chow, 2011; Ghirarduzzi, et al., 2010; Hanna, et al., 2014; and Ju, et al., 2013). When individuals refuse treatment, their QOL may diminish depending on the severity of the disease. A common motif that was existent throughout the search, is patients with IDC who have undergone treatment

often go into remission and have an improved QOL (Bianchi, et al., 2012; Dennis & Chow, 2011; Fenicia, et al., 2014; Ghirarduzzi et al., 2010; Hanna, et al., 2014, and Ju, et al., 2013). The general objective of treatment is to mitigate pain, improve QOL, and provide physical and mental health support to patients and their families. Deciding on the most beneficial treatment for all cancer patients is critical to the patient's overall state of being. Patients experience different symptoms from the treatments, so physicians must approach the differences with urgency.

PHYSICAL AND MENTAL HEALTH

Metastatic breast cancer patients experience a decline in their QOL after being diagnosed. Mental disorders occur in as many as 50 percent of patients with cancer. Many patients experience a low state or significant state anxiety as a result of treatment (Asevedo, et al., 2013; and Ghirarduzzi, et al., 2010). These studies suggest that this anxiety correlates with items considered to potentially influence the QOL, such as the patient's outlook on life, chemotherapy induced side effects, breast symptoms, and stresses caused by physical changes. Metastatic breast cancer patients also experience lower QOL because of the physical impact the cancer has on the patient's body. For example, a woman that undergoes a double mastectomy is physically limited because she loses her ability to breastfeed. "Breast cancer is the most frequent neoplasm that can metastasize to the head and neck region" (Kechagias, et al., 2012). A patient whose cancer metastasizes to another region experiences additional complications to those already associated to the original diagnosis.

LIMITATIONS

A consideration is the minimal number of articles which have a focus in the area of quality of life. This showed a significant need for research specific to this area to assist in both the treatment of IDC, but also to increase the QOL for those with the disease. Seemingly, QOL is a factor which goes hand-in-hand with treatment success.

CONCLUSION

It was important to examine the QOL factors in relation to IDC treatment because it allows healthcare professionals to gain insight on the impact these factors have on the overall well-being of the patients. The literature review identified various positive and negative QOL factors through a systematic review of current articles in two scholarly databases. This information may be useful for physicians, other clinicians, and healthcare administrators attempting to improve the QOL of IDC patients.

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